

Exploration and Preparation of SDSS Quasar Data for Deep Learning Applications

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The Sloan Digital Sky Survey (SDSS) has provided one of the most extensive and publicly available spectroscopic datasets in astronomy, including millions of spectra for stars, galaxies, and quasars. Quasars - extremely luminous active galactic nuclei powered by supermassive black holes - are of particular interest due to their brightness and visibility across vast cosmological distances. Identifying quasars and accurately determining their redshifts from spectroscopic data is a foundational task in extragalactic astronomy and cosmology.

Traditional methods for redshift estimation and classification rely on human expertise or template fitting, which, while effective, are time-consuming and scale poorly with the growing volume of astronomical data. Recent advancements in deep learning have shown promise in automating and enhancing this process. One notable example is QuasarNET, a deep convolutional neural network designed to identify quasars and estimate their redshifts by mimicking the decision-making process of expert astronomers. It does this by detecting specific emission line features in the spectra and using their observed wavelengths to infer redshifts.

The goal of my internship at SARL, NCGSA is to build a simplified deep learning model that can perform classification and basic redshift estimation on SDSS quasar spectra. While not attempting to fully replicate the architectural complexity of QuasarNET, this work draws inspiration from its core idea: treating classification and redshift estimation as a feature detection problem. The model aims to take pre-processed flux data as input and learn to distinguish quasars from other spectral classes, while optionally estimating the redshift based on the positions of emission lines.

To support this, spectra were downloaded from the SDSS database and pre-processed to be suitable for training. This includes normalization, wavelength alignment, and noise filtering. The model architecture will be kept minimal to ensure interpretability and allow rapid experimentation, especially for educational and proof-of-concept purposes. If time permits, a partial implementation of QuasarNET may be attempted to explore how its architecture achieves high accuracy and robustness in real-world SDSS datasets.

Ultimately, this project serves as a bridge between astrophysical data analysis and modern machine learning techniques, showcasing the potential of neural networks to transform large-scale spectroscopic analysis in astronomy.

1 AGNs and Galaxies

AGNs, or Active Galactic Nuclei, are the extremely bright, compact regions at the centers of some galaxies, powered by a supermassive black hole actively consuming surrounding matter. This process generates intense energy across the electromagnetic spectrum, often outshining the entire host galaxy, and can also launch powerful jets and outflows. The Active Galactic Nuclei (AGN) have been divided into two broad categories based on their radio detection.

- **Radio-Loud AGNs** are active galactic nuclei with very strong radio emission, powered by large, relativistic jets that shoot out from the central supermassive black hole. These jets produce intense radiation, making the AGN bright in radio wavelengths. They are usually found in massive elliptical galaxies.
- **Radio-Quiet AGNs** have weak radio emission and lack powerful jets. Most of their energy comes from the accretion disk and surrounding regions, with radio

waves mainly from small-scale outflows or star formation. They are often found in spiral galaxies and make up the majority of AGNs.

Galaxy:

- Collection of stars, gas, dust, and dark matter bound by gravity.
- The central black hole is inactive.
- Light mainly comes from stars.

AGN:

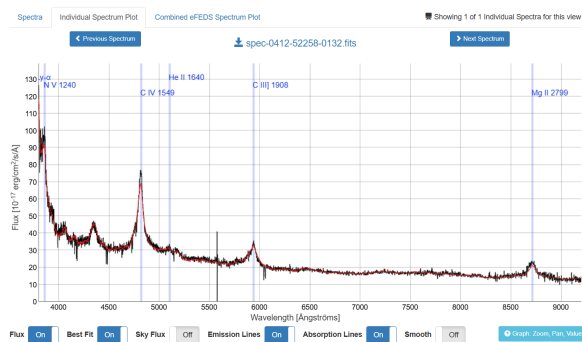
- Also, a Galaxy
- Central supermassive blackhole is actively emitting.
- Light mainly comes from powerful jets.

Aspect	Radio-Loud Galaxy	Radio-Quiet Galaxy
Radio Flux	Very high due to strong synchrotron emission from jets	Low, dominated by optical/UV emission
Jets	Powerful, large-scale relativistic jets	Weak or absent
Orientation Effect	Strongly affects appearance (e.g., blazar if jet points toward us)	Minor effect, mostly optical differences
Examples	Radio galaxies, blazars, radio-loud quasars	Seyfert galaxies, most quasars

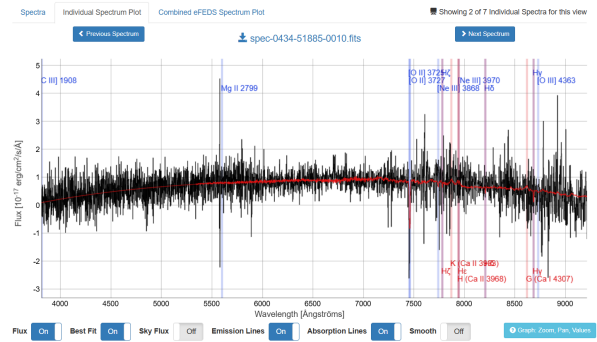
Table 1: Difference between Radio-loud and Radio-quiet galaxies

2 Quasars

The **Quasars** are extremely bright active galaxies powered by a supermassive black hole at their center that is rapidly consuming surrounding material. As matter spirals into the black hole, friction in the accretion disk heats it to very high temperatures, causing intense radiation and making the quasar outshine its entire galaxy.



(a) Spectrum of a Quasar



(b) Spectrum of a Galaxy

Figure 1: Comparison of a Quasar and a Galaxy.

Quasars are among the most luminous objects in the universe, visible across billions of light-years. While some material falls into the black hole, part of it is ejected outward as powerful jets at nearly the speed of light, especially in radio-loud quasars.

3 What is SDSS

The Sloan Digital Sky Survey (SDSS) is one of the most ambitious and influential astronomical surveys ever conducted. It began operations in the year 2000 with the primary goal of creating the most detailed three-dimensional map of the universe. Utilizing a dedicated 2.5-meter telescope at Apache Point Observatory in New Mexico, SDSS has systematically captured optical images and spectra of millions of celestial objects, including stars, galaxies, quasars, and other cosmic structures.

The main objective of SDSS has been to improve our understanding of the structure, formation, and evolution of the universe by collecting high-quality data on a massive scale. Over the years, SDSS has produced some of the largest and most precise maps of the cosmos, significantly advancing research in cosmology, galaxy formation, and large-scale cosmic structures.

In addition to its primary survey, SDSS has given rise to several specialized offshoot surveys. The MaNGA Stellar Library (MaSTAR) focuses on obtaining high-quality stellar spectra to study the physical properties of stars in detail. The Baryon Oscillation Spectroscopic Survey (BOSS) was designed to measure the large-scale distribution of galaxies and quasars, with the aim of studying dark energy through baryon acoustic oscillations. These specialized programs have broadened the scientific impact of SDSS, allowing it to address a diverse range of astronomical questions.

3.1 Data releases

A **data release** in the Sloan Digital Sky Survey (SDSS) refers to the official public release of data collected by the survey over a specific period. These releases include imaging and spectroscopic data, sky coverage maps, object catalogs, and other useful resources for the astronomical community. Each subsequent release expands on previous ones by adding new data, improving calibration techniques, and enhancing data processing algorithms. Researchers worldwide rely on these data releases for studies in cosmology, galaxy formation, quasar research, and other areas of astrophysics.

3.1.1 Early Data Release (EDR)

The SDSS began its operations on May 9, 1998 (Loveday et al., 1998). However, the first public data release, known as the **Early Data Release (EDR)**, was made in 2001. This release included imaging of the north Galactic cap and smaller regions of the south Galactic cap.

- **Sky Coverage:** 462 square degrees of imaging data.
- **Spectra:** 54,008 spectra recorded, including galaxies, quasars, and stars (Stoughton et al., 2002).
- **Quasar Catalog:** 3,814 quasars were cataloged with a full width at half-maximum (FWHM) larger than 1000 km s^{-1} and luminosity greater than -23 (Schneider et al., 2002).

The EDR served as a test and demonstration of SDSS capabilities, laying the groundwork for future larger data releases.

3.1.2 First Data Release (DR1)

Building upon the success of EDR, SDSS expanded its observational footprint and made its first major public release, the **First Data Release (DR1)**, in June 2003. This marked a significant step forward in terms of both area covered and the number of spectra available.

- **Sky Coverage:**

- Imaging data: 2,099 square degrees
- Spectroscopic data: 1,360 square degrees

- **Spectra:** 186,240 total spectra of galaxies, quasars, and stars (Abazajian et al., 2003).
- **Quasar Catalog:** 16,713 quasars identified (Schneider et al., 2003). The luminosity threshold was updated to -22 with the same FWHM criterion of $> 1000 \text{ km s}^{-1}$.

DR1 established SDSS as a leading survey, providing a robust dataset for the global research community.

3.1.3 Ninth Data Release (DR9)

The **Ninth Data Release (DR9)** marked another milestone in SDSS, being the first release to include data from the **Baryon Oscillation Spectroscopic Survey (BOSS)**. BOSS was designed to map large-scale structures in the universe and study dark energy through measurements of baryon acoustic oscillations.

- **Sky Coverage:** 14,555 square degrees of imaging data.
- **Catalog Objects:** 932,891,133 objects.
- **Spectra:**
 - Galaxy spectra: 1,457,002
 - Quasar spectra: 228,468
 - Stellar spectra: 668,054
- **Special Updates:**
 - First release of BOSS spectra covering over 3,300 square degrees.
 - Improved astrometry based on DR8 calibrations.
 - Updated stellar parameters using the SEGUE Stellar Parameter Pipeline (SSPP).

DR9 combined all previous SDSS data and introduced higher-resolution data products, significantly expanding the scope of research possible with SDSS.

3.1.4 Eighteenth Data Release (DR18)

The **Eighteenth Data Release (DR18)** marks the beginning of the fifth phase of the Sloan Digital Sky Survey (**SDSS-V**). This release represents a major step forward for the survey, as it introduces new scientific goals and advanced instrumentation, building upon the legacy of previous phases of SDSS. DR18 is described in detail in the official data release paper by Almeida et al. (2023).

DR18 is primarily focused on three main scientific programs, also called **Mappers**:

- **Milky Way Mapper (MWM)**: Dedicated to mapping the structure and stellar populations of the Milky Way galaxy in unprecedented detail.
- **Black Hole Mapper (BHM)**: Focused on studying the growth and evolution of supermassive black holes across cosmic time.
- **Local Volume Mapper (LVM)**: Aiming to create high-resolution maps of nearby galaxies and interstellar medium structures.

This release contains extensive **targeting information** for the MWM and BHM programs, including:

- Input catalogs and selection functions for various scientific goals.
- Black Hole Mapper eFEDS spectra, part of the SPIDERS science program.
- A value-added catalog (VAC) of redshifts and classifications for all DR18 eFEDS targets.

In addition, DR18 includes approximately **25,000 new spectra** and supplemental information for X-ray sources identified by the **eROSITA** telescope in its eFEDS field. It also introduces several updates to the SDSS software pipelines and provides previews of improvements expected in DR19.

3.2 Significance of DR18 for Our Study

For our research, DR18 was the primary source of data. It provided high-quality, up to date spectroscopic and imaging data essential for our analysis of galaxies and quasars. The targeting catalogs and precise spectroscopic measurements in DR18 allowed for reliable selection and characterization of the objects included in our study.

- **Sky Coverage**: Expanded coverage across key survey regions using SDSS-V instrumentation.
- **New Spectra**: ~25,000 newly added spectra, including targets from the eROSITA eFEDS field.
- **Special Features**: Introduction of value-added catalogs with enhanced redshift and classification data.

3.3 Working with SDSS

To carry out our study, we used spectroscopic data from the Sloan Digital Sky Survey (SDSS) database, specifically from **Data Release 18 (DR18)**. The SDSS database provides a dedicated interface to search and download spectroscopic data, available at <https://dr18.sdss.org/optical/spectrum/search>. This platform allows users to apply specific filters to refine their search and extract only the data relevant to their research.

Each SDSS spectrum is uniquely identified using three parameters:

- **Plate Number:** The spectroscopic plate used to capture the observation.
- **MJD (Modified Julian Date):** The exact date of the observation.
- **Fiber ID:** The specific fiber on the plate that was assigned to the object.

The combination of these three values (*plate-MJD-fiber*) serves as a unique key to locate and reference any spectrum within the SDSS archive.

Data Selection Process: When downloading our data, we carefully selected filters to ensure that only the desired spectra were retrieved. These selections were made through the web interface provided by SDSS DR18:

- **Spectrograph Option:** Selected *SDSS* only. The interface offers three choices: *SDSS*, *BOSS*, or *Both*. Since our study is based on SDSS data, we chose SDSS exclusively to avoid mixing datasets from different surveys.
- **Class Option:** Selected *QSO (Quasi-Stellar Object)*. Other available classes were *GALAXY* and *STAR*, but our focus was on quasars.
- **Redshift Range:** Set between **1.0 and 3.0**. This range was chosen to include quasars at moderate to high redshifts, ensuring that we capture their key spectral features while avoiding extremely low-redshift or excessively distant objects.
- **Additional Parameters:** Plate, MJD, and Fiber ID fields were left open during the search to retrieve all relevant objects within the chosen class and redshift range.

After setting these filters, we clicked on the **Search** button, which returned a table of results. This table included important columns such as:

- **specObjID** – a unique identifier for each spectrum.
- **z** – measured redshift of the object.
- **zerr** – redshift error.
- **S/N** – signal-to-noise ratio, indicating the quality of the spectrum.
- **Plot** – an option to visualize the spectrum before downloading.
- **File** – a direct link to download the FITS file.

Downloading Spectra: For individual spectra, the direct **File** option was used to download FITS files manually. However, when working with a larger sample, we utilized the **wget** command-line tool for bulk downloads.

The SDSS interface provides a list of direct download links, which can be saved into a text file, for example, `spectra_list.txt`. We then executed the following command:

```
wget -i spectra_list.txt -P ./sdss_spectra/
```

Where:

- `-i spectra_list.txt` reads the list of file URLs.
- `-P ./sdss_spectra/` specifies the local directory where the spectra are saved.

Additional useful `wget` options included:

- `--continue`: Resumes interrupted downloads.
- `-r`: Enables recursive downloading of entire directories.
- `-np`: Prevents `wget` from downloading parent directories.

3.3.1 The SDSS FITS File

The Sloan Digital Sky Survey (SDSS) data are provided in the **FITS (Flexible Image Transport System)** file format, which is the standard format for storing astronomical data. Each SDSS spectral FITS file consists of multiple **Header-Data Units (HDUs)**. Each HDU contains either a data table or metadata describing the observations. The structure of an SDSS spectrum FITS file is as follows:

HDU 0: Primary Header The first HDU contains only a header with general metadata about the observation and no data array. This metadata includes information such as the object’s position and observation details. Some important keywords in this header are:

- **PLUG_RA, PLUG_DEC** – The right ascension and declination of the object.
- **THING_ID, FIBERID** – Unique identifiers linking the observation to the SDSS database.
- **NEXP, EXPID** – Information about the number of exposures contributing to the final spectrum.

These details are primarily copied from the parent `spPlate` file with some modifications to match the specific object.

HDU 1: Main Science Spectrum (Binary Table HDU) This is the most important HDU and contains the actual scientific data for the spectrum. Each row in this table represents one pixel of the spectrum and includes several essential columns:

- **loglam** – The logarithm (base 10) of the wavelength in Angstroms. This is used to compute the wavelength for each pixel using:

$$\lambda = 10^{\text{loglam}}$$

- **flux** – The calibrated flux density, measured in units of $10^{-17} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ \AA}^{-1}$. This represents the amount of energy received per unit area, time, and wavelength.
- **ivar** – The inverse variance of the flux, which is used to estimate the uncertainty in the flux measurement.
- **mask / or_mask** – Bitmasks that indicate problematic pixels, such as those affected by cosmic rays or poor sky subtraction.
- **model** – The best-fit model spectrum used for redshift estimation.
- **sky** – The contribution of the sky background that was subtracted from the final spectrum.

For our work, we primarily relied on the **loglam** and **flux** columns. The **loglam** column was used to reconstruct the wavelength scale, while the **flux** column provided the spectral intensity at each wavelength, forming the basis of our analysis.

HDU 2: Summary Metadata (Binary Table HDU) This HDU provides classification and survey-level information for the object. It is directly copied from the SDSS **spAll** catalog and contains essential identification and measurement details. Some of the key columns include:

- **SURVEY** – Indicates the survey in which the observation was made (e.g., SDSS, BOSS).
- **INSTRUMENT** – Specifies whether the data was collected using the SDSS or BOSS spectrograph.
- **SPECOBJID** – A unique identifier based on the combination of plate, MJD, and fiber ID.
- **CLASS** – Spectroscopic classification of the object (e.g., STAR, GALAXY, QSO).
- **Z, Z_ERR** – The estimated spectroscopic redshift and its associated uncertainty.

In our study, the most important information extracted from this HDU was its measured **redshift (Z)**. Which we used for our wavelength correction, i.e., from red-shifted wavelength to rest-frame wavelength.

HDU 3: Emission-Line Fitting Results This HDU contains the results of emission-line fitting, where Gaussian profiles are fitted to prominent spectral lines after subtracting a stellar absorption-line template. While this data can provide information about line strengths and widths, it was not directly used in our analysis and was therefore not included in our workflow.

4 Data Augmentation

For this project, a total of 15,955 FITS files were downloaded from SDSS DR18, containing quasar spectra within the selected redshift range of 1.0 to 3.0. This dataset formed the initial pool of raw data to be processed and analyzed.

To ensure data quality, spectra were manually reviewed and labeled as either **Good** or **Bad**:

- **Good Spectra:** Clean, smooth spectra with identifiable quasar features.
- **Bad Spectra:** Spectra with missing data, abnormal spikes, excessive noise, or flat profiles lacking clear quasar signatures.

Initially, over **1,500** spectra were classified manually by visually inspecting each file. Due to time constraints, this number was reduced as the project progressed, and only a subset of the total data was used for the final stages of the analysis. This step ensured that only high-quality data was included, preventing unreliable results due to corrupted or noisy inputs.

4.1 Pipeline Development and Processing

To handle the large volume of data efficiently, a custom data processing pipeline was developed iteratively. The pipeline automated repetitive tasks and ensured consistency across all datasets. Its main stages included:

1. **Conversion of FITS to NPY format:** Each FITS file was read and converted into a NumPy (`.npy`) format. This significantly simplified data handling and allowed for efficient loading into Python scripts for analysis.
2. **Plotting and Wavelength Correction:** The pipeline generated plots for each spectrum, making it easier to visually identify anomalies. Wavelength corrections were also applied to standardize the spectra before further processing.
3. **Automated Multi-format Output:** Once complete, the pipeline processed over **5,100** FITS files and generated outputs in three formats:
 - `.npy` files for machine learning and computational analysis.
 - `.png` images for visual inspection of spectra.
 - `.pdf` files for documentation and report-ready figures.
4. **Spectral Smoothing:** To reduce noise and highlight essential spectral features, a **Savitzky–Golay** (`savgol`) filter was applied. This smoothing technique improved interpretability by minimizing random fluctuations while preserving key quasar emission lines.

4.2 Testing on Data Subsets

Since manually reviewing the full dataset of 15,955 spectra was not feasible within the project timeline, we progressively worked with smaller subsets of the data:

- The initial subset of approximately 1,500 spectra was fully classified into good and bad categories.
- Subsequent stages used even smaller, refined subsets for pipeline testing and validation.
- This iterative approach allowed us to debug the pipeline and refine classification methods without processing the entire dataset at once.

This stepwise refinement ensured that the final analysis was based on high-quality, well-understood data while keeping the workload manageable and aligned with the project's time constraints.

4.3 Classification of Spectra

The downloaded spectra were manually reviewed and categorized into two main classes: **Good Spectra** and **Bad Spectra**. This classification was essential to ensure that the analysis was based on clean and interpretable data.

Good Spectra: Spectra that were smooth, continuous, and exhibited identifiable quasar features were labeled as *Good*. These spectra were suitable for further processing and scientific interpretation.

Bad Spectra: Spectra were classified as *Bad* if they exhibited any of the following issues:

- Missing or incomplete data.
- Abnormal spikes or sudden discontinuities.
- Broad Absorption Line (BAL) quasars with complex, distorted profiles.
- Excessive noise or very low signal-to-noise ratio (S/N).
- Numerous downward dips or upward spikes are inconsistent with quasar patterns.
- Flat or straight spectra lacking any distinguishable quasar trends.

4.4 Working with Python

Python is a high-level, open-source programming language known for its simplicity and versatility. Created by Guido van Rossum in 1991, it has become one of the most widely used languages in scientific computing due to its extensive ecosystem of libraries such as NumPy, SciPy, Astropy, Matplotlib, and Pandas. In scientific data analysis, Python enables efficient handling of large datasets, data visualization, and automation of workflows. Its readability and strong community support make it ideal for research applications, including **Astronomy and Astrophysics**, where large volumes of data need to be processed, analyzed, and interpreted.

4.5 Data Analysis

The initial data analysis was carried out using a step-by-step pipeline implemented in Python scripts and custom programs. The overall workflow is summarized below:

- **Reading FITS Files:** All downloaded FITS files were first converted into NumPy (.npy) format. This conversion simplified the handling of large datasets and allowed for faster reading and plotting of spectra.
- **Accessing Information:** Key information such as `flux`, `loglam` (logarithm of wavelength), and `redshift` (z) was extracted from the FITS file headers. A new column for the **rest-frame wavelength** was calculated using the formula:

$$\lambda_{rest} = \frac{\lambda_{observed}}{1 + z}$$

This step corrected the observed wavelength to the quasar’s rest frame, ensuring consistency across all spectra.

- Example plots of raw spectra were generated to visually confirm proper extraction and correction.

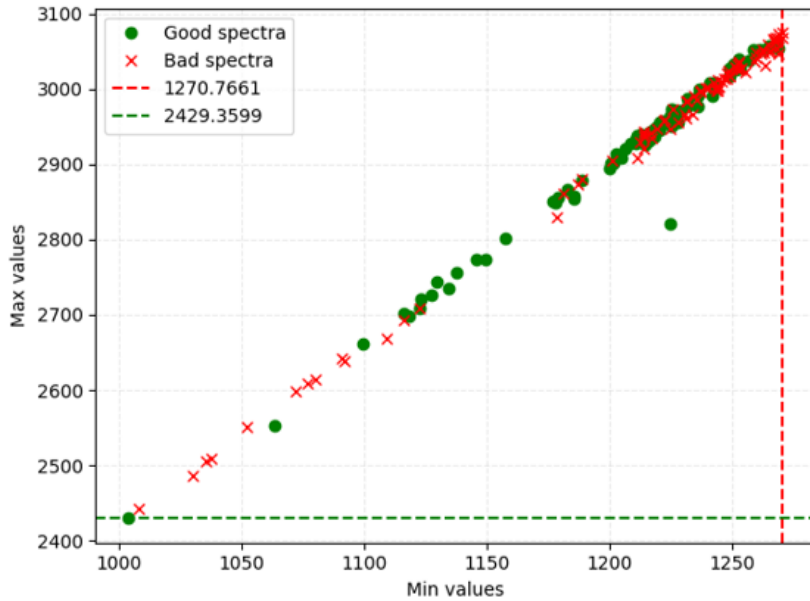


Figure 2: The min/max values and boundaries for combined GOOD/BAD spectra.

- **Wavelength Discrepancy Correction:** Different spectra in the dataset covered slightly different wavelength ranges. For instance, one spectrum might start at 1090 Å and end at 3000 Å, while another might span from 1200 Å to 2400 Å. This caused unequal numbers of data points, which posed a problem for machine learning models. To standardize the data, we:

1. Found the **maximum of all minimum wavelengths** across the dataset.
2. Found the **minimum of all maximum wavelengths** across the dataset.

All spectra were then trimmed to this common range, resulting in uniform data dimensions suitable for modeling. **Figure 2** shows the maximum and minimum of the max/min values of good/bad spectra, see figure 2.

- **Noise Reduction Using the Savitzky–Golay Filter:** The raw spectral data contained significant noise, making visual classification difficult. A **Savitzky–Golay (savgol)** filter was applied with:
 - Window length = 10
 - Polynomial order = 2

This smoothed the spectra while preserving important spectral features.

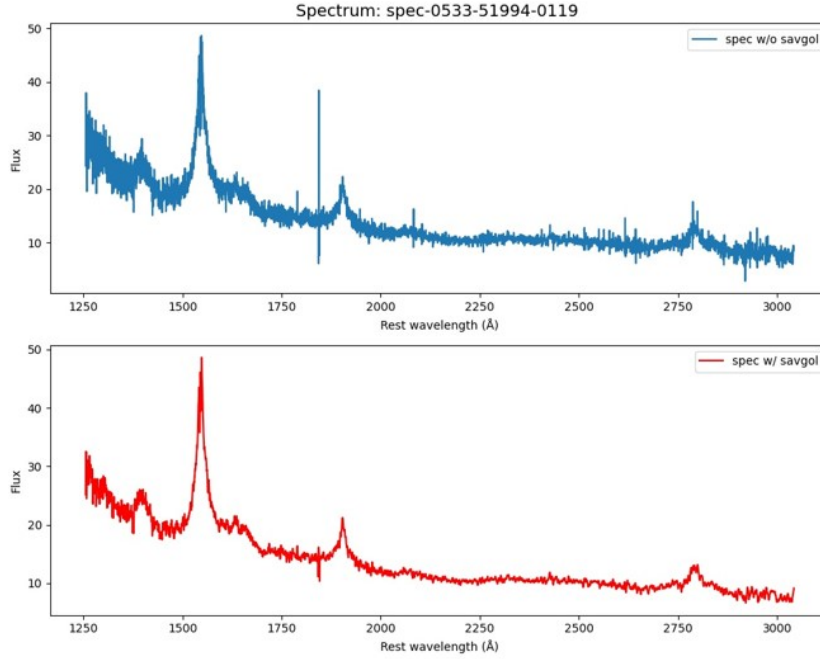


Figure 3: Spectrum w/o savgol and w savgol

- **Identifying Key Emission Lines:** Quasar spectra are characterized by distinct emission lines. For our dataset, which spanned approximately 1200–2400 Å, we focused on three major lines:
 - O IV ~ 1400 Å
 - C IV ~ 1550 Å
 - C III ~ 1900 Å

These lines were chosen because they are strong indicators of quasar properties and were consistently present within our wavelength range.

- **Power-Law Fitting:** Quasar spectra typically shows a decaying continuum that follows a power-law trend. To extract a meaningful feature for machine learning classification, we fitted each spectrum with a power-law model and calculated the **power-law index**.

- The power-law index was observed to differ between good and bad spectra, making it a useful distinguishing parameter.
- **Visualization and Analysis:** Several plots were generated to illustrate:
 - Example of a **good spectrum** with clear quasar features.



Figure 4: This spectrum is smooth with a low S/N ratio, and our emission lines are clearly visible

- Examples of **bad spectra** with missing or noisy features.

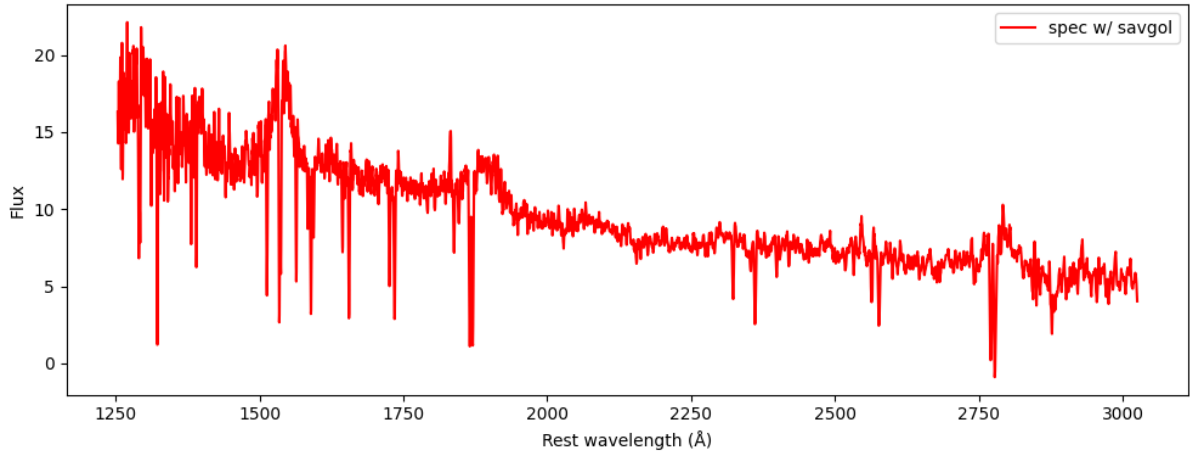


Figure 5: Spectrum clearly shows too much S/N ratio and dips.

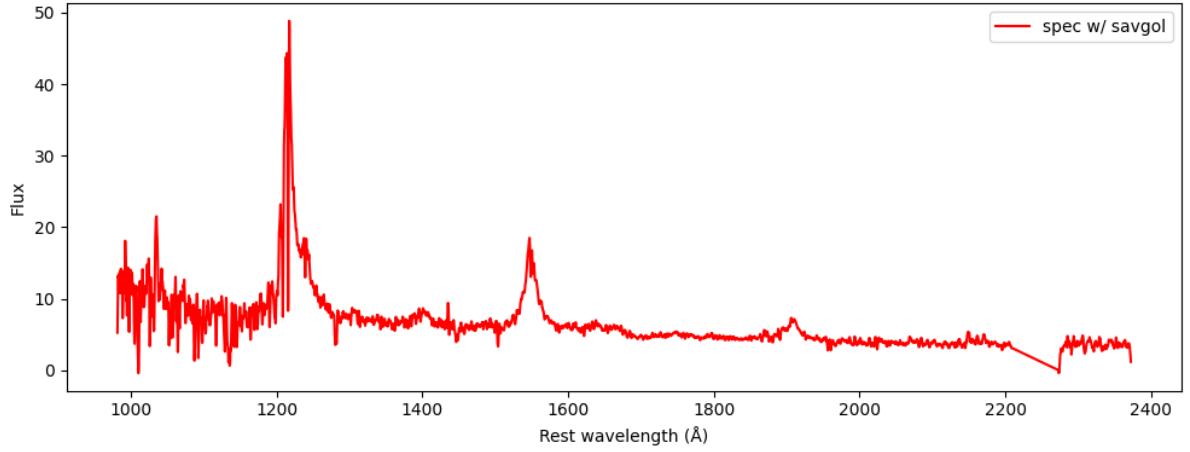


Figure 6: Spectrum shows missing data after 2200 Å

- The differences in power-law index between good and bad spectra.

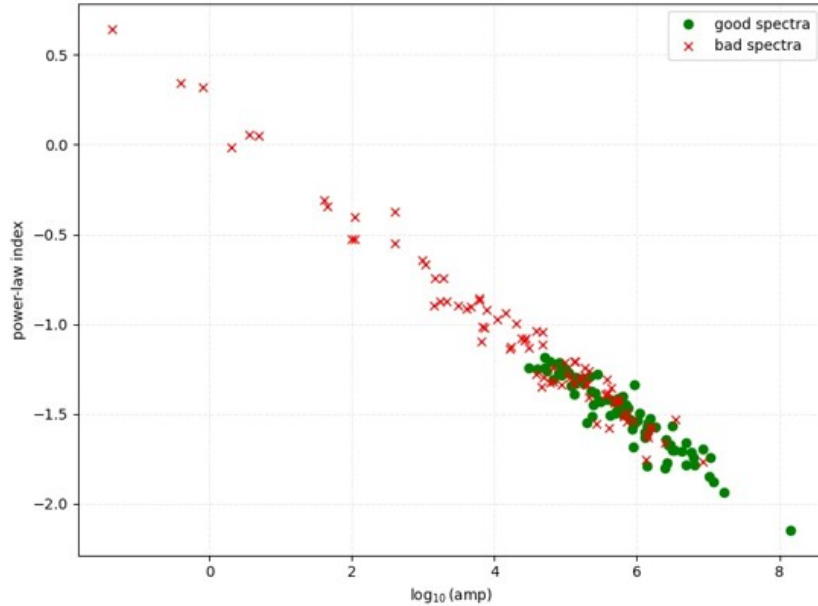


Figure 7: This plot shows the power law index and the amplitude of our spectra.

These visualizations provided qualitative confirmation of the effectiveness of our pipeline.

5 Conclusion

Through this exercise, I gained a comprehensive understanding of the Sloan Digital Sky Survey (SDSS), its significance in modern astronomy, and how its vast datasets can be used for research. I learned how SDSS data is downloaded, processed, and analyzed, as well as how machine learning can assist in handling large volumes of astronomical data that would otherwise be challenging to process manually. This project also highlighted the importance of programming for data visualization, enabling me to better understand spectral features and trends.

I developed skills in research and literature review, learned how to approach problems creatively, and explored how small changes - such as adjusting the wavelength range - can significantly affect the results and interpretation of data. I also understood the nuances involved in visually and programmatically classifying quasar spectra (QSO), which is essential for building reliable machine learning models.

During the project, I identified several key classification pointers that differentiate good spectra from bad ones. These include:

- Spectrum shape and overall structure
- Power law index (indicating decay trends in the QSO spectrum)
- Normalized Flux
- Signal-to-noise ratio (SNR)
- Emission line detection
- Final label (good or bad)

The data preprocessing and cleaning stages are now complete. The next step in this exercise will be to develop and train a deep learning model that can automatically classify the spectra using these parameters. This will allow for scaling the classification process to much larger datasets and reduce the need for manual inspection.

Overall, this experience enhanced my understanding of astronomical data analysis and gave me valuable insights into applying computational methods and machine learning in astrophysics research.

5.1 Future Discussion

Well, time was short, and so is life. While we managed to complete the data preprocessing and cleaning, we could not fully implement the deep learning classification model within the given timeframe. In the future, we'll aim to continue this work by developing and training a deep learning model to automatically classify spectra. This will allow for faster, more accurate, and scalable analysis, taking our project beyond manual classification and into the realm of modern machine learning applications in astrophysics.